

Yinghao (Peter) Guan

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TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript (ES6+), SQL, Shell Scripting

AI & Research: LLM Integration (Gemini, Perplexity), Prompt Engineering, XAI (Explainable AI), ASR Pipelines

Backend & Systems: Node.js, REST APIs, WebSockets, Firebase, Linux/Unix, Git, Maven

Frontend & Tools: JavaFX, HTML5/CSS3, LaTeX, ANTLR, Next.js

HACKATHON & RESEARCH EXPERIENCE

Realibuddy — Real-time Conversational Fact-Checking Assistant

Nov 2025

Backend Lead / HackCC (Best AI/ML Award - 1st Place among 29 teams)

- Spearheaded the development of an AI-driven system to combat misinformation in live conversations.
- Architected a high-concurrency streaming pipeline using **WebSockets** and **Deepgram ASR** for sub-second latency.
- Integrated **Gemini** and **Perplexity** APIs to perform real-time RAG (Retrieval-Augmented Generation) for claim verification.
- Engineered fallback logic and "human-in-the-loop" UI notifications to ensure high accuracy and user trust.
- Developed and deployed the full-stack prototype within a 14-hour intensive sprint.

HCI Research — Explainable AI (XAI) for Emotional Support LLMs

Jul 2025 – Present

Undergraduate Researcher / Santa Monica, CA

- Investigating the impact of transparency on user trust and empathy in AI-mediated mental health support.
- Designing and executing an **A/B user study** comparing standard LLM responses vs. XAI-enhanced outputs.
- Leveraging sentiment analysis and emotion detection to build more "human-centric" conversational agents.
- Analyzing quantitative survey data and qualitative interviews; currently co-authoring a paper for submission.

MyMD — Custom Markup DSL & Compiler

Aug 2025 – Present

Project Lead / Java, ANTLR, Visitor Pattern

- Developing a specialized DSL to bridge the gap between Markdown's simplicity and LaTeX's technical precision.
- Implementing a full compiler frontend using **ANTLR4**, including lexing, parsing, and AST generation.
- Building a modular backend that transforms AST nodes into Pandoc-compatible formats for academic publishing.
- Applied software design patterns (Visitor, Strategy) to ensure the compiler is extensible for new target languages.

ADDITIONAL PROJECTS

Behavioral Modeling of Excessive Trading

Jan 2025 – Apr 2025

Data Analyst / Python, R (PLS-SEM)

- Conducted statistical analysis on behavioral biases in financial markets using PLS-SEM modeling.
- Processed and visualized complex datasets to identify correlations between psychological traits and trading frequency.

EDUCATION

Santa Monica College

2025 – Present

Associate in Science: Computer Science | GPA: 4.0

Relevant Coursework: C++, Calculus I.

University of Edinburgh

2023 – 2024

Coursework in Computer Science

Selected: Functional Programming, Computer Systems, Software Engineering & Professional Practice.